

Faster Speech-LLaMA with Multi-token Prediction

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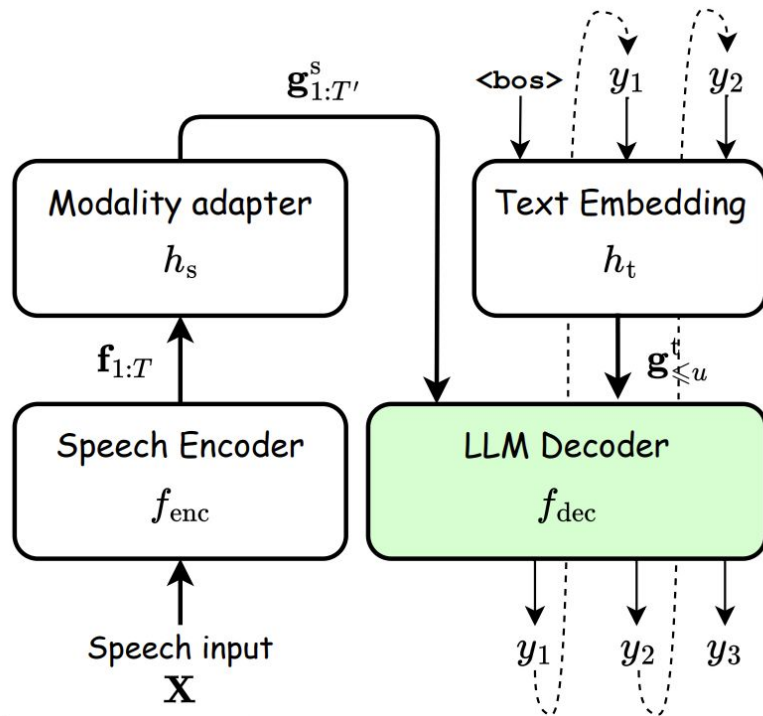


Overview

- Background: ASR with LLMs (Speech-LLaMA)
- Problem: slow inference due to autoregressive generation
- Solution: multi-token prediction
- Model architectures
- Inference methods
- Experimental results
- Conclusion

Background: Speech-LLaMA

- Speech-LLaMA combines a speech encoder with a pre-trained LLM-based decoder, usually through a modality adapter such frame subsampling.
- The encoder is usually pre-trained using CTC loss.
- The joint model is fine-tuned for ASR using cross-entropy loss.



Problem

- Due to the sequential nature of autoregressive decoding, the decoder requires as many steps as the number of tokens in the ASR output.
- Since the LLM decoder is usually large in Speech-LLaMA, this process incurs a high computation cost and latency.
- In practice, this manifests as a high real time factor (RTF), i.e., the time required to transcribe per second of audio.

Solution: multi-token prediction

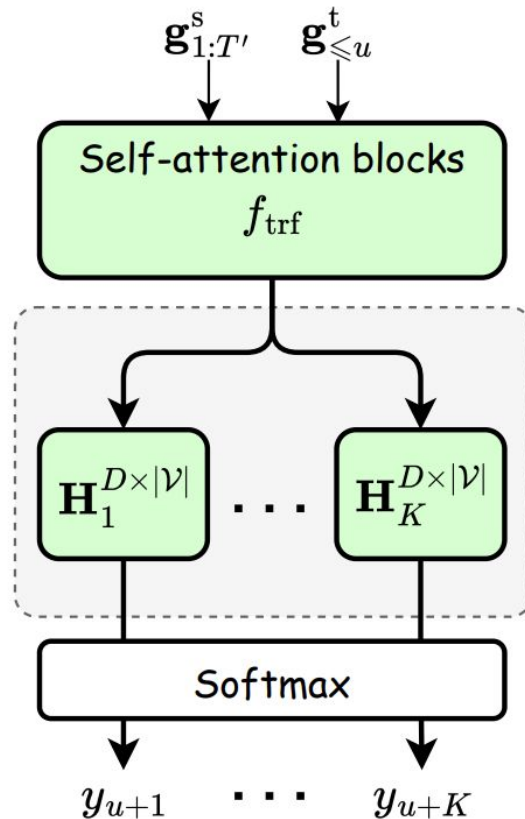
- Instead of predicting 1 token every time step, we want to predict K tokens.
 - If the total number of predicted tokens is N , this only requires N/K steps to finish transcription.
 - In the ideal case, this strategy reduces the number of decoder calls by K times.
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- This is a kind of speculative decoding strategy for LLMs [1], [2].

[1] Gloeckle, Fabian et al. “Better & Faster Large Language Models via Multi-token Prediction.” (2024).

[2] Cai, Tianle et al. “Medusa: Simple LLM Inference Acceleration Framework with Multiple Decoding Heads.” (2024).

Model architectures

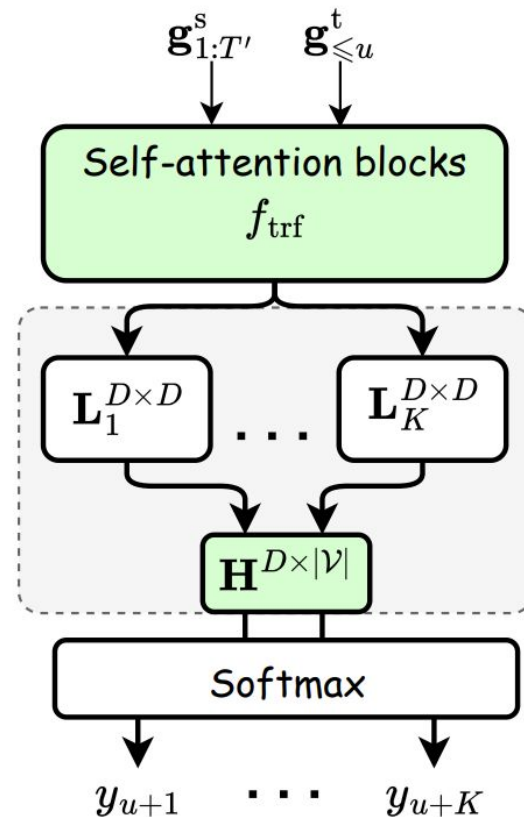
- **Option (a)**
- We have K linear projection heads, each predicting one output token.
- However, modern LLMs have large vocabularies, so this may prohibitively increase the model size.
- Example: 125M parameter LLM decoder with vocab size 32K $\rightarrow$ 225M with $K=4$.



(a)

Model architectures

- **Option (b)**
- Use K projection heads in the latent space.
- Example: 125M parameter LLM decoder with vocab size 32K $\rightarrow$ 130M with K=4.



(b)

Inference methods

1. Verification-based:

- a. “Predict” the next set of K tokens.
- b. “Verify” that this sequence would be identical to the one generated using autoregressive decoding.
- c. “Accept” the largest subsequence that satisfies the verification step.

2. Threshold-based:

- a. “Predict” the next set of K tokens.
- b. Retain the tokens with probability higher than some threshold.
- c. Allows better control over decoding speed versus accuracy.

Experimental setup

LibriSpeech

- 960h training data in English

Multi-lingual ASR

- ~700k hours of data in English, French, Italian, German, and Spanish.
- Includes MLS, CommonVoice, VoxPopuli, and in-house data.

Model

- 125M distilled LLaMA used as decoder
- VGG-Conformer encoder; 100M for LS, 200M for multi-lingual ASR

Results

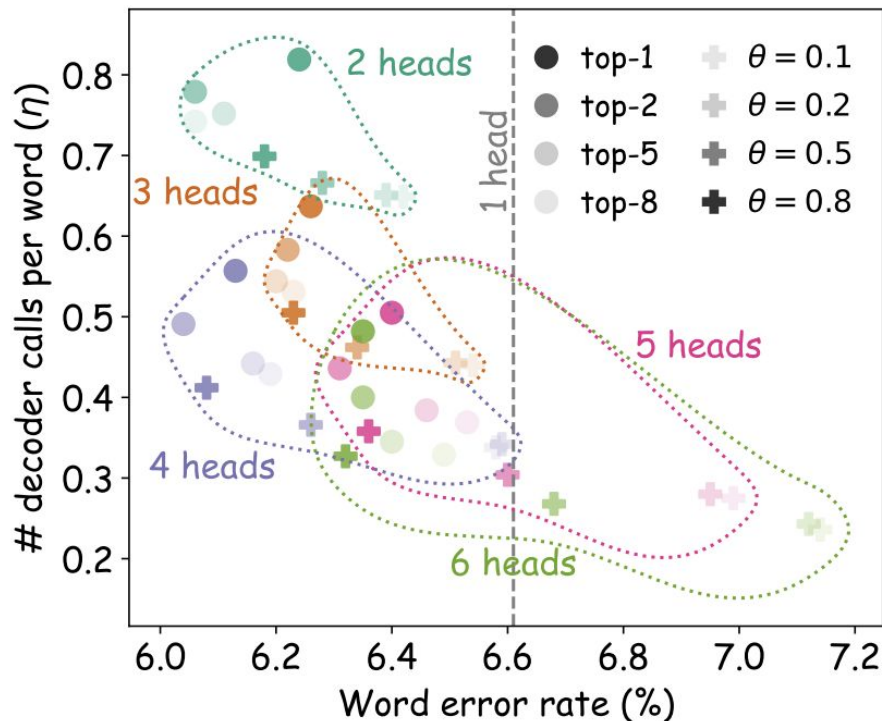
Latent head model performs better than linear projection.

TABLE I: Comparison of multi-token prediction architectures and decoding methods in terms of ASR performance (% WER) and the decoding speed, measured by η (13) and the decoder RTF.

Model	Decoding method	test-clean			test-other		
		WER	η	RTF	WER	η	RTF
Speech-LLaMA	—	4.3	1.25	0.79	6.9	1.26	0.79
Projection (4-head)	top-1	4.1	0.54	1.15	6.4	0.56	1.16
	top-5	4.0	0.43	0.81	6.4	0.45	0.85
	$\theta = 0.8$	4.0	0.38	0.63	6.3	0.42	0.61
Latent (4-head)	top-1	3.8	0.53	0.91	6.4	0.55	0.86
	top-5	3.7	0.42	0.63	6.3	0.44	0.63
	$\theta = 0.8$	3.7	0.38	0.47	6.2	0.41	0.45

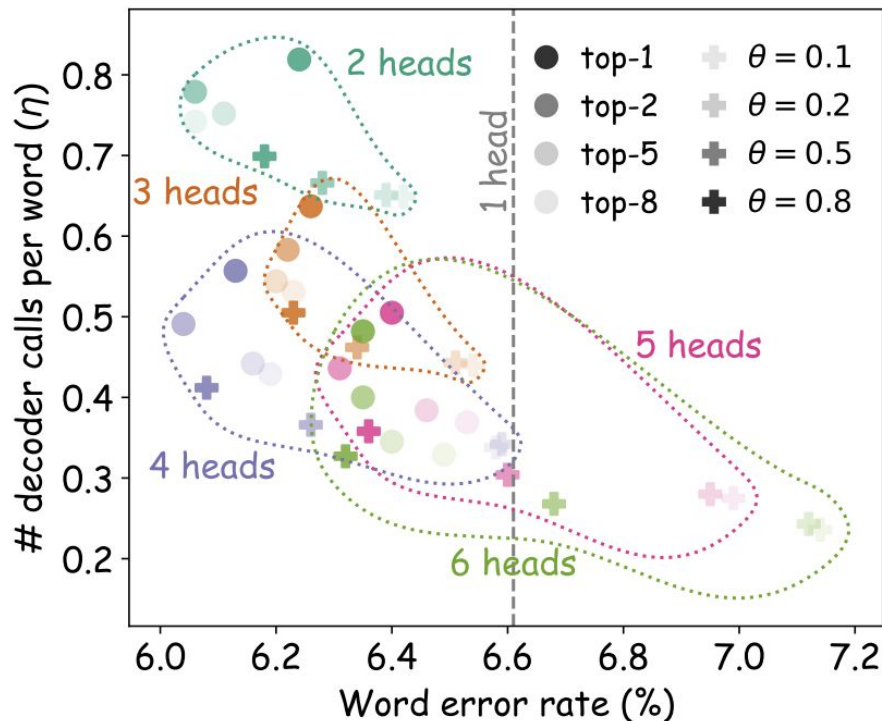
Results

WER is consistent upto K=4, but degrades on increasing K further.



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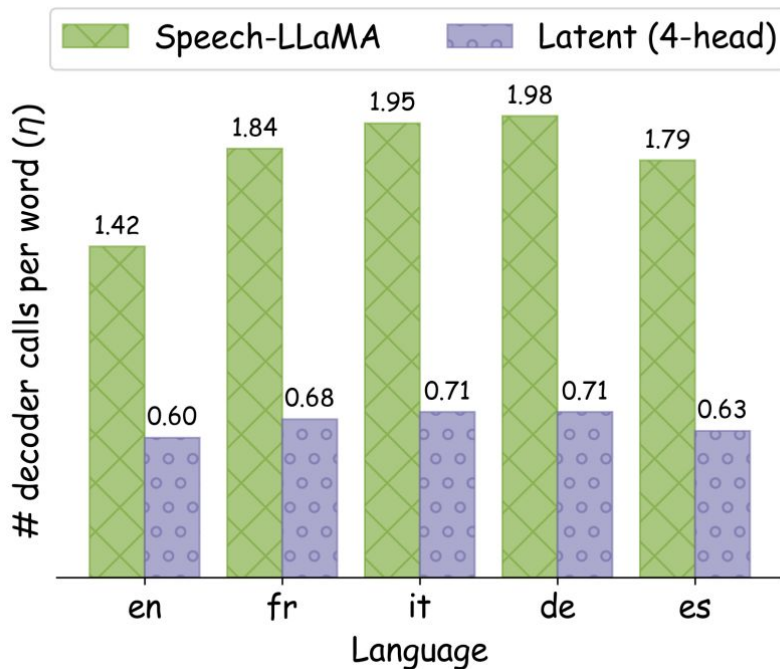
~4% relative WER degradation for multi-lingual ASR.

TABLE II: Comparison of 4-head latent multi-token prediction model using threshold-based inference ($\theta = 0.8$) vs. baseline Speech-LLaMA (in % WER) on multi-lingual benchmarks.

Model	Common Voice					VoxPopuli					MLS				
	en	fr	it	de	es	en	fr	it	de	es	en	fr	it	de	es
Speech-LLaMA	12.1	9.3	6.4	6.5	6.4	6.9	8.5	13.6	9.5	7.2	6.0	4.1	8.3	5.2	3.5
+ MWER	11.1	9.0	6.1	6.2	5.9	6.9	8.5	13.3	9.0	7.0	5.8	4.0	8.2	5.1	3.3
Latent (4-head)	12.5	9.8	6.9	7.0	6.7	7.3	9.0	13.7	9.6	7.2	6.3	4.3	8.2	5.4	3.7
+ MWER	12.0	9.7	6.8	6.9	6.5	7.3	8.6	13.7	9.5	7.2	6.2	4.3	8.2	5.4	3.6

Results

Number of decoder calls reduces significantly for all languages (~62% on average).



Conclusion

- We proposed a method to speed-up ASR decoding with Speech-LLaMA by predicting multiple tokens in the same decoding step.
- In addition to the well known linear projection strategy, we proposed a latent projection technique to prevent large increase in number of model parameters.
- The proposed model reduces the number of decoder calls by 3.2x while maintaining or improving WER performance.